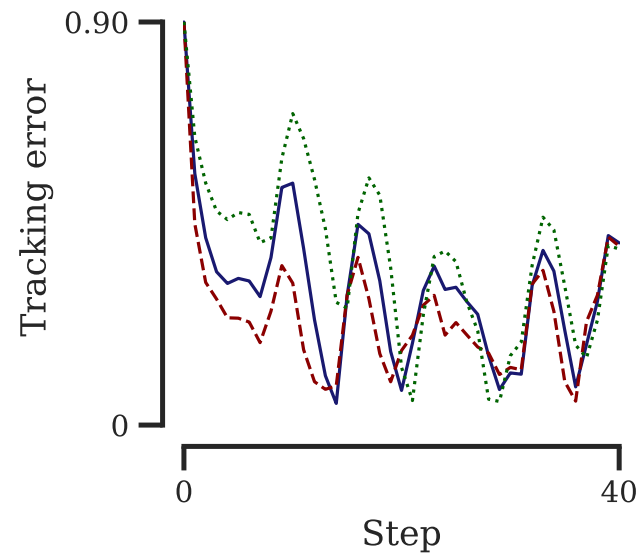


Synthetic controller comparison

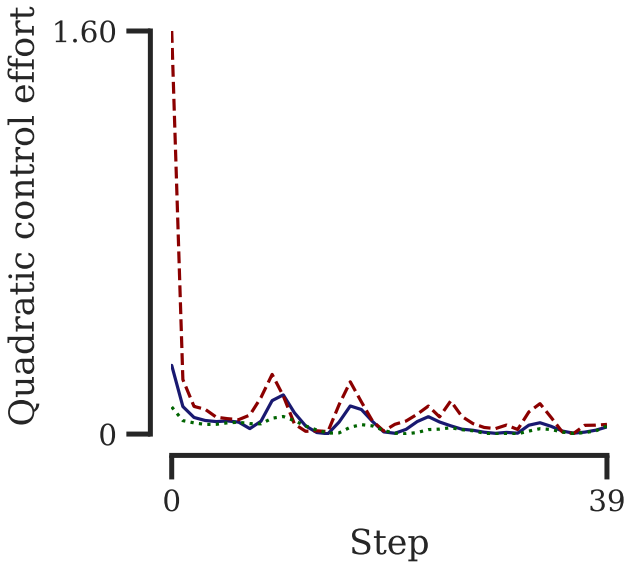
Two states · 40 steps · matched conditions · seed 2151

— LQR - - H_∞ ... PID

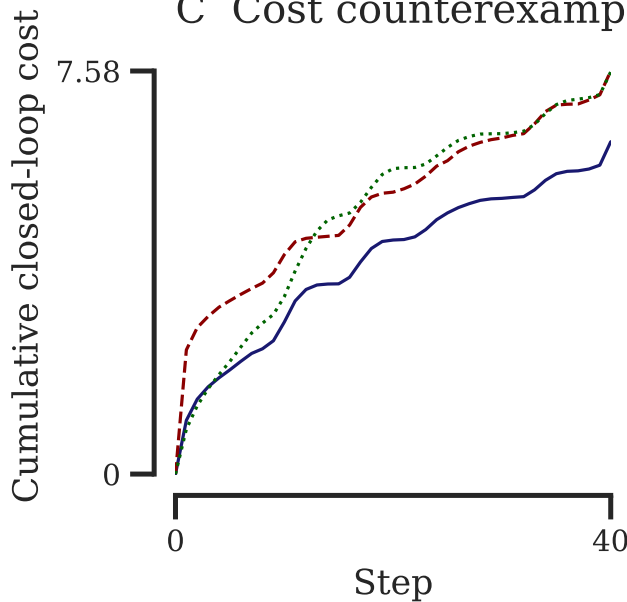
A Tracking error



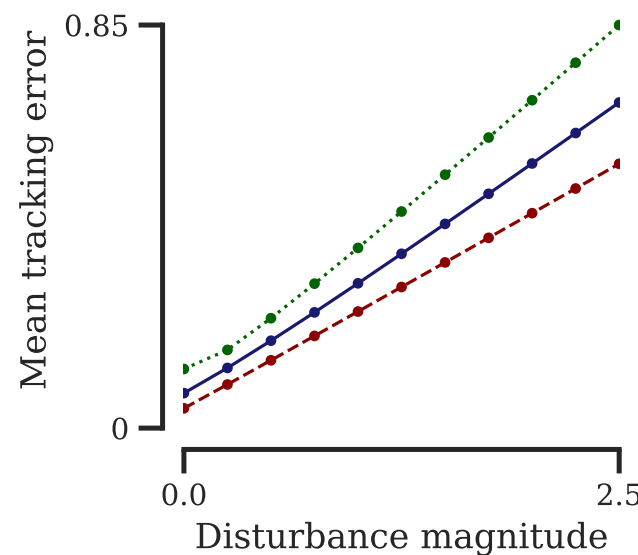
B Control effort



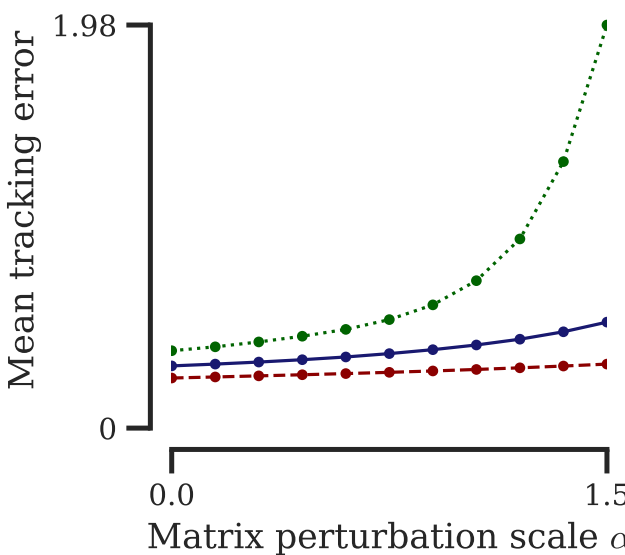
C Cost counterexample



D Disturbance sensitivity



E Model-uncertainty sensitivity



F Rollout summary

	LQR	H_∞	PID
Mean error	0.305	0.246	0.380
Maximum error	0.901	0.901	0.901
Effort J_u	1.90	4.49	1.03
Total cost J	6.25	7.58	7.56
Failure flag	no	no	no

H_∞ synthesis: $\gamma^* = 0.648$

One fixed disturbance realization; no error bars. Sensitivity sweeps use one disturbance shape and one matrix direction. Synthetic error-coordinate system; control effort excludes reference feedforward. Failure flag: maximum error ≥ 10 or nonfinite.